

YANG LIU

Department of Economics, 8000 York Road ■ Towson, MD 21252
■ (732) 485-7011 ■ yangliu@towson.edu

EMPLOYMENT

Assistant Professor, Department of Economics, Towson University, Towson, MD

August 2024 –

EDUCATION

Department of Economics, Rutgers University, New Brunswick, NJ

May 2024

Ph.D. in Econometrics and Quantitative Economics

PhD Dissertation: Methods for Forecasting Using High Dimension and High Frequency Data

Committee: Norman R. Swanson (Chair), John Chao, John Landon-Lane, Xiye Yang

Robert H. Smith School of Business, University of Maryland, College Park, MD

May 2018

Masters of Quantitative Finance: Asset Management and Risk Management

Capital University of Economics and Business, Beijing, China

June 2016

Bachelor of Management: International Human Resources Management

FIELDS OF INTEREST

- **Applied Econometrics, Financial Economics, Machine Learning, Big Data Forecasting, Asset Pricing**
- **Concentration: high-dimensional data, time series data, real-time data, factor model, variable selection**

PUBLICATIONS

“An Assessment of the Marginal Predictive Content of Economic Uncertainty Indexes and Business Conditions Predictors”, with Norman R. Swanson, *International Journal of Forecasting*, Volume 40, Issue 4, October–December 2024, Pages 1391-1409

“Liquidity Spillover between Exchange-Traded Funds: Variations across News Regimes”, with Yongchen Zhao, *Journal of Risk and Financial Management*, 2024, 17(9), 391

WORKING PAPERS

Consistent Estimation, Variable Selection, and Forecasting in Factor-Augmented VAR Models (R&R at *Quantitative Economics*)

- (with John C. Chao, Kaiwen Qiu and Norman R. Swanson)

■ *Abstract:* We introduce a completely consistent method for variable selection with high dimensional datasets. The method is presented in a framework where latent factors are estimated for the purpose of dimension reduction and is meant to serve as a complement to extant methods. We argue that the method is of particular interest in empirical settings where there may be many irrelevant predictor variables. The reason for this is that situations where there are “too many” irrelevant variables can lead to inconsistent factor estimates. Interestingly, our method yields a consistent estimate of the number of such irrelevant variables, which can aid the applied practitioner in assessing the strength of the underlying factor structure for a particular application. We also show that when factors constructed using our variable selection method are inputted into an estimated factor augmented vector autoregressive (FAVAR) model for the purpose of forecasting, the associated conditional mean forecasting functions can be consistently estimated. Monte Carlo results are presented indicating that the variable selection method performs well in finite samples. In an empirical illustration, we simulate a real-time FAVAR forecasting environment, and show that using our selection method yields forecasts that are sometimes better than those associated with simply using all variables or using thresholding when constructing factors,

even in small samples, suggesting that our method may offer a useful complement to extant variable selection methods when estimating factors.

WORK IN PROGRESS

Can Stock Characteristics Lead the Forecasting for High-frequency Volatilities?

TEACHING EXPERIENCE

Assistant Professor, Department of Economics, Towson University, Towson, MD Fall 2024 –

- **Macroeconomic Principles (Econ 202) – Undergraduate course** (Fall 2024 – Now)
- **Computational Economics (Econ 431) – Undergraduate course** (Fall 2024, Fall 2025)
- **Computational Economics (Econ 631) – Master’s Program course** (Fall 2024, Fall 2025)
- **Introduction to Mathematical Economics (Econ 451) – Undergraduate course** (Spring 2025)
- **Microeconomic Principles (Econ 201) – Undergraduate course** (Spring 2026)

Lecturer, Department of Economics, Rutgers University, New Brunswick, NJ Summer 2021, 2022, 2023

- **Intermediate Microeconomics – Undergraduate course** (Summer 2021)
- **Econometrics – Undergraduate course** (Summer 2022)
- Prepared teaching notes, assignments, two exams and two review sessions for each of above two courses.
- Received the final overall evaluation as 4/5 for both of above two courses.
- **Introduction to R** (one-day workshop for the **first-year Ph.D. students** in Summer 2022 and Summer 2023).

Teaching Assistant, Department of Economics, Rutgers University, New Brunswick, NJ 2019—2023

- Econometrics II (Ph.D. students), Advanced Statistics (Ph.D. students), Big Data Forecasting, Economics of Taxation, Capital Markets, Economics of Taxation, Introduction to Microeconomics, Introduction to Macroeconomics

WORK EXPERIENCE

Internship (BNY Mellon)

- Data Modeler, Economic Forecasting Group, BNY Mellon, New York City, NY June 2022—Oct. 2022

- Re-estimated different models using updated new data for several financial assets or macroeconomic variables (such as ABS spreads variables, covered bonds and VIX) in order to find new drivers to improve their forecasting performances.
- Conducted CCAR stress tests for multiple macroeconomic variables in various aspects and recorded each of them in a complete document for model validation team.
- For each model: examined the coefficient stability, fitted in crisis era with calibration sample, conducted out-of-sample forecasting, plotted model fit condition in graph and conducted residual tests.
- Made a presentation for model development progresses in the weekly team meeting and explained different models in economic concepts.
- Prepared slides for financial market news update and macroeconomic overview in the quarterly company conference.

Research Assistant, Rutgers University, New Brunswick, NJ

R project - Developing A New R Package “LavaCvxr” March 2021—June 2021

- Developed and published “LavaCvxr” package in R using CVXR for Lava method, which is introduced in Chernozhukov et al. (2017).
- Paper for Lava method: Chernozhukov, V., Hansen, C., & Liao, Y. (2017). A lava attack on the recovery of sums of dense and sparse signals. *Annals of Statistics*, 45(1), 39-76.

Research Assistant, University of Maryland, College Park, MD

SAS project - Manufacturing Sector Oct. 2016—Dec. 2016

- Constructed linear regression with fixed effect to predict revenue growth for hundreds of companies within manufacturing sector to see the influences from firm level variables and macroeconomic variables.

- Did winsorization for outliers, used stepwise method and VIF results to select final regressors in the model.
- Did two sample T test to compare the results between public companies and private companies; checked heteroscedasticity.

MATLAB project - Consumer Nondurables Sector and S&P 500 Index

Oct. 2016—Dec. 2016

- Forecast return of consumer nondurables sector by using financial and macroeconomic variables.
- Used Dickey Fuller test to check data stationary; did backward discard and forward selection for regressors selection.
- Durbin Watson test: check first-order-autocorrelation; White test: check heteroscedasticity; Jarque-Bera test: check normality.
- Forecast 1-step ahead variance for S&P 500 series by using GARCH(1,1) of 50 years daily data.

CONFERENCES, WORKSHOPS AND PRESENTATIONS

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| ■ 2025 NBER-NSF Time Series Conference | Sep. 2025 |
| ■ Econometrics Seminar, Rutgers University | Nov. 2024 |
| ■ Trade & Development Seminar, George Washington University | Oct. 2024 |
| ■ The 7 th International Conference on Econometrics and Statistics (EcoSta 2024) | July 2024 |
| ■ 2023 Annual Meeting of the International Society for Data Science and Analytics (ISDSA) | July 2023 |
| ■ 2023 The 37th Annual Conference of the Pennsylvania Economic Association (PEA) | June 2023 |
| ■ 2023 The Chinese Economists Society China Annual Conference (CES) | June 2023 |

JOURNAL PEER REVIEWER

- *International Journal of Forecasting* (2), *Open Economies Review* (1), *Journal of Risk and Financial Management* (1)

TECHNICAL SKILLS

- **Coding:** R, MATLAB, Python (Pandas, NumPy, Matplotlib, Scikit-Learn), SQL, SAS
- **Machine Learning:** Lasso, PCA, Random Forest, Gradient Boosting, Neural Network, K-means clustering, SVM
- **Passed CFA Level I Test**
- English (fluent), Mandarin (native)

HONORS AND AWARDS

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| ■ Alfred S. Eichner Economics Prize, Rutgers University | Spring 2023 |
| ■ Richard Lock Endowed Fund for Economics Award, Rutgers University | Spring 2023 |
| ■ Dorothy Rinaldi Fellowship, Rutgers University | Spring 2023 |
| ■ MMI Market Structure Research Fellowship, Modern Market Initiative | Spring 2023 |
| ■ Teaching Assistantship, Rutgers University | 2019-2022 |
| ■ 2nd award in 2014 China Undergraduate Mathematical Contest in Modeling (CUMCM) | 2014 |